

Research Statement — Chris Smith: Machine Learning in Biology

Deep neural networks—a kind of machine learning—have outperformed traditional statistical models in a growing number of cases. In fact, in some situations they have opened the door for entirely new kinds of analyses. Part of the reason neural networks are exciting, is their ability to extract useful information from *complex* molecular, environmental, and imaging data sources, in a partly automated fashion. Also, deep learning is a *flexible* modeling framework, with rapidly developing theory and continual addition of new modeling tricks; this means that powerful models can be tailored for addressing urgent biodiversity or health applications, or longstanding questions in biology. I am exploring the potential of machine learning in biology, and am also fascinated by the computer science and methods development components involved with building new machine learning tools. In addition to developing methods, I apply them to real data, contribute to community software resources, and collaborate with others to apply my methods to their study systems.

I. Machine learning for genomic estimation of spatial population features

Contributions. Obtaining estimates for where individuals are (population density) and information about movement (dispersal rate) are important for applications in ecology and evolutionary biology, for example, studying climate change response and adaptation; monitoring endangered, invasive, or disease vector species; and tracking the spread of adaptive alleles. One potential strategy for efficiently monitoring these demographic features is using spatial genomic data (Figure 1a): Genomes are a rich data source and thus fewer specimens and potentially non-invasive sampling techniques may be used. However, current genetics-based methods have constraints that prevent their use in many species, leaving a critical gap in our methods toolbox. To address this need, I developed machine learning tools for estimating dispersal rate from population genetic data (Smith et al. 2023, *Genetics*; Smith and Kern 2023, *BMC Bioinformatics*). These tools train on simulated data and can be used with single nucleotide polymorphisms (SNPs), making it possible to infer dispersal rate for non-model organisms with limited genomic resources. I applied my tool, **disperseNN**, to 12 publicly available datasets and found reasonable estimates for most species. I also developed a machine learning approach for estimating *maps* of population density and dispersal across the landscape (Figure 1b; Smith et al. 2024, *Molecular Ecology Resources*). This method, **mapNN**, is useful for identifying barriers to migration, population-dense areas, and source-sink dynamics, and I applied it to publicly available North American grey wolf data.

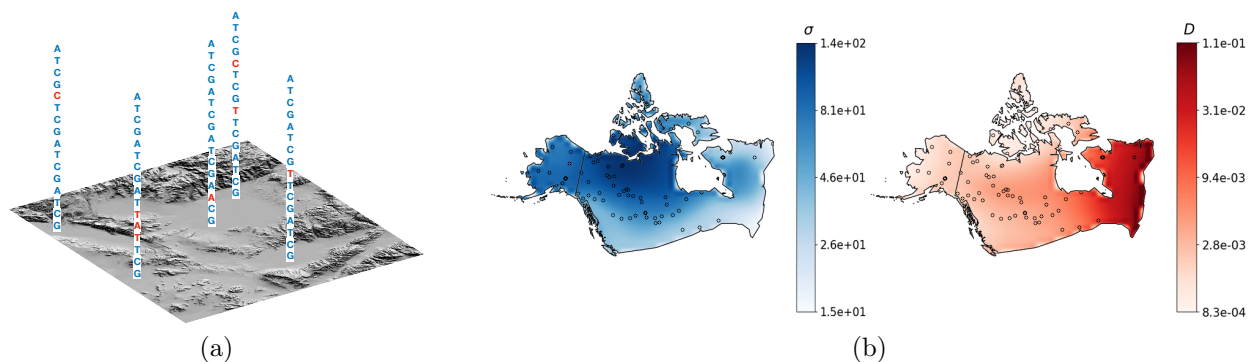


Figure 1: (a) Diagram of geographically-indexed DNA sequences—the input for my inference methods. (b) Estimated maps of dispersal rate and population density in North American grey wolves.

Next Steps. I am planning an R35 representing an extension to my F32 award “Deep learning for spatial population genetics”. The general motivation for this work is described above, but the NIH proposal will be angled around studying (i) mosquitoes—a malaria vector and the world’s most deadly animal—, and (ii) horseshoe bats (*Rhinolophus*; 106 known species)—host to a diversity of viruses, including ones relevant to human health such as hantaviruses, flaviviruses, ortheoviruses, and the likely progenitor of SARS-Cov-2. The connectedness of the modern era means more frequent pandemics, which makes efficient wildlife monitoring extremely relevant for future public health. Obtaining data on population density and dispersal will be vital for population surveillance, prioritizing insecticide targets, planning gene drive efforts, understanding evolutionary dynamics of insecticide resistance, and modeling virus evolution.

II. Demographic history inference

Barn swallows (*Hirundo rustica*) are a model study system in ecology and evolutionary biology, however the population history of the six barn swallow subspecies is not well resolved. In collaboration with Dr. Rebecca Safran, I hypothesized and compared different models of population history for barn swallows using an approximate Bayesian computation (ABC) framework combined with random forests, a type of supervised machine learning (Smith et al. 2018, *Molecular Ecology*). As one of the first studies to implement ABC using genome-scale data, my analysis estimated that the timing of subspecies divergence was an order of magnitude more recent than previous estimates (Figure 2a). During this project, I noticed that ABC methods did not take full advantage of the information provided by whole-genome sequencing: specifically, haplotype information. To address this limitation, I modified the program `msABC` to calculate *haplotype* summary statistics that improved the inference of recent demographic parameters (Smith and Flaxman 2020, *Molecular Ecology Resources*).

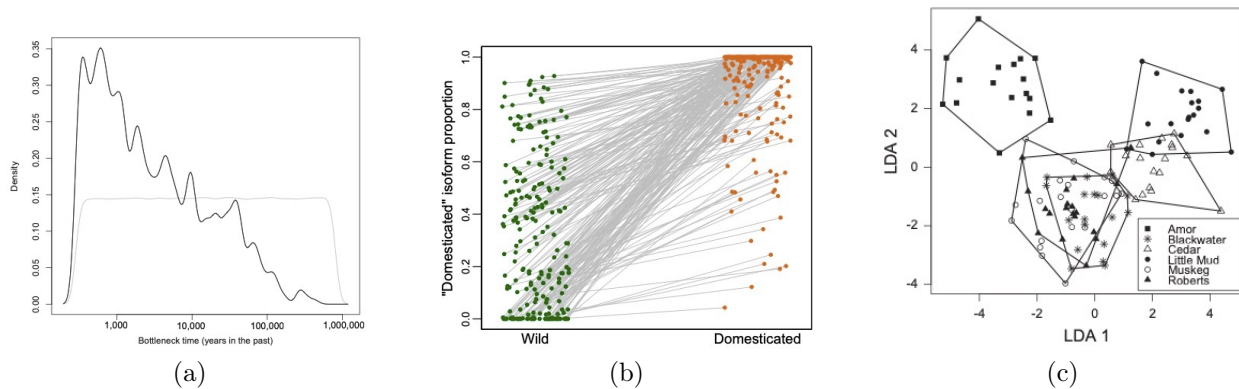


Figure 2: (a) Posterior density for barn swallow bottleneck time (black) and prior distribution (grey). (b) Sunflower genes with differentiated mRNA splicing. Points show the mean proportion of the “domesticated” isoform in each parental type. (c) The first two axes from a linear discriminant analysis using principal coordinates as input that describe variation in gut microbiota among individuals (points) and populations (convex hulls).

III. Speciation genomics

One longstanding question in evolutionary biology is: What is the molecular mechanism of population divergence? With Dr. Nolan Kane’s lab, I found differentially-spliced mRNA isoforms between wild and domesticated sunflowers (*Helianthus annuus*; Figure 2b), suggesting that changes in mRNA splicing may be a target of natural selection (Smith et al. 2018, *PNAS*). My study is one of the first to explore transcriptome-wide splicing differentiation between closely related, non-human populations. Next, I discovered incorrectly-spliced transcripts in hybrid sunflowers (Smith et al. 2021, *Evolution*). The erroneous transcripts negatively affected seedling growth rate, and many were regulated by multiple alleles with nonadditive interactions. These findings suggest that splicing errors—and population divergence—could be the molecular manifestation of many small-effect genetic incompatibilities.

IV. Microbial ecology

To explain differences in gut microbial communities we must determine how the processes regulating community assembly differ among hosts and affect microbiota composition. With Dr. Daniel Bolnick, I studied natural populations of threespine stickleback (*Gasterosteus aculeatus*) to identify pathways of microbial colonization of the gastrointestinal tract. Using high-throughput 16S rRNA sequencing, I found that sticklebacks from different lakes harbored different gut microbes (Figure 2c), and after controlling for food-associated and environmental microbes the gut microbiota appeared to be partly inherited (Smith et al. 2015, *The ISME Journal*). Currently, I am developing a machine learning framework for analyzing which environmental factors control soil microbes’ uptake of atmospheric methane. I look forward to building and deploying machine learning tools for making additional discoveries in each of the aforementioned research areas in biology.